



PROCESS BOOK

BEYOND THE PITCH

The Data Story of Women's International Football

1956 – 2025

From a 50-year ban to 2 billion viewers — told through 11,177 matches, 10 D3 visualizations, and 70 years of data.

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Team DVP

01 Overview & The Original Idea



1.1 Introduction

We have always been told that women's football is a minor sport. A footnote. Something that happens in the shadow of the men's game. But what if the data tells a completely different story? What if you look closely at the numbers and find a sport that was *literally banned* in England until 1971, and is now watched by 2 billion people and offers \$110 million in World Cup prize money?

"*Beyond the Pitch*" was born from this exact curiosity. We wanted to move beyond the standard sports statistics websites that show tables and match results. Our goal was to create a guided, interactive data journalism piece that investigates the full arc of women's football: from suppression to global phenomenon. We wanted users to not just see the data, but to *feel* the scale of that transformation.

We structured our story around a central question: **how did a banned sport become a global movement, and what does the data reveal about every step of that journey?**

1.2 The dataset

To build a comprehensive narrative, we used a single authoritative source from Kaggle: **Women's International Football Results**, which contains three CSV files covering 69 years of international matches.

- **matches_clean.csv** — 11,177 matches from 1956 to 2025, with date, teams, scores, tournament type, venue, and geographic region. 247 different teams/nations across 117 tournament types. Zero missing values.
- **goalscorers.csv** — 2,795 individual goal records with scorer name, minute, penalty flag, and own-goal flag. 1,055 unique scorers across multiple decades.
- **shootouts.csv** — 134 penalty shootout results since 1984, with first-shooter outcome data.

On top of these core files, we manually compiled supplementary data from FIFA and Wikipedia: World Cup attendance figures per edition (1991–2023), prize money per edition for both women's and men's tournaments, and WSL league attendance. These small tables were built directly in our preprocessing notebook to contextualize the economic story.

This dataset gave us everything we needed: the scale of global growth, the economics of the sport, and the individual human stories behind the matches.

1.3 Target Audience

Our target audience is anyone who follows football, or anyone curious about how a marginalized sport became a global movement. We intentionally designed the experience to be intuitive for a general audience: a user with no statistical background should be able to scroll through all ten chapters and understand every insight without needing to read documentation.

02 The Journey: From Concept to Reality



2.1 Initial Brainstorming

When we first conceptualized this project during Milestone 1, we asked three central questions:

1. How did the sport grow from a handful of European nations to a truly global game?
2. What does the economic data reveal about how women's football has been valued — or undervalued — compared to the men's game?
3. Who are the legends, and what do the individual match patterns tell us about the sport itself?

Initially, we imagined a highly technical dashboard. We envisioned multiple side-by-side panels, complex cross-filtering systems, and a dedicated section for pitch-level tactical heatmaps using StatsBomb's open event data. We had ambitious plans for showing individual passes, shots, and tackles from the 2019 and 2023 World Cups.

However, as we progressed through the design phase, we realized that presenting users with a wall of simultaneous charts is overwhelming. It forces them to make their own analytical choices before they understand the context, and context is everything in this story.

2.2 The Scrollytelling Pivot

We decided to adopt a **scrollytelling architecture**, we built a custom scroll-driven navigation system with ten sequential chapters. We wanted to control the narrative flow, guiding the user step by step through the story before arriving at the individual details.

Instead of letting users get lost in data, we introduce concepts in a deliberate order: first **History** (how the sport grew), then **Geography** (where it spread), then **Power** (which teams dominate) and so on... Each chapter builds emotional and analytical context for the next.

This decision fundamentally changed the architecture of the project. Rather than a single interactive dashboard, we built what is effectively a ten-chapter interactive article — one that can be read linearly, but also explored non-linearly through the sticky navigation menu.

03 Evolution of Visualizations

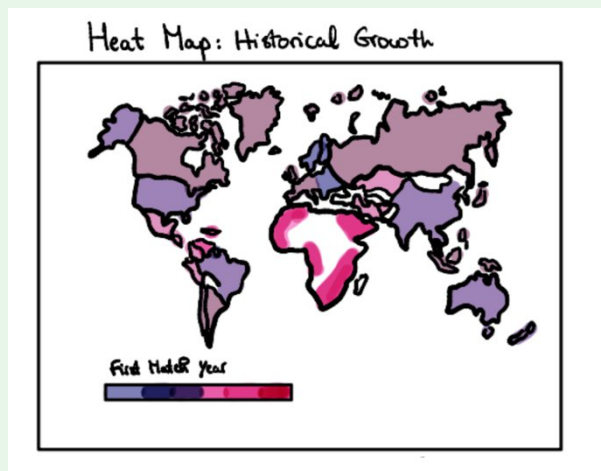


3.1 The World Map

Our first sketch for the geographic view was a **static choropleth map** — a world map where each country was colored by its "First Match Year" using a continuous blue-to-pink gradient. Earlier nations (1950s–1970s) appeared in deep blue; later entrants (2000s–2020s) appeared in pink or rose. The idea was to give an immediate visual impression of which regions adopted the sport first. The static snapshot had a fundamental limitation: it showed the *result* of 69 years of growth without conveying the *process*.

We replaced the single continuous color scale with a **three-state animation system**: rose for a country's debut year, gold for active participation, and grey for inactive or not yet entered. Driven by a year slider from 1956 to 2025, the user can now watch nations light up one by one across decades. A play button animates the spread automatically, turning 69 years of history into a two-minute visual story. We also changed the color encoding deliberately: the debut rose is a moment of arrival, not a position on a scale, the new system communicates events rather than values.

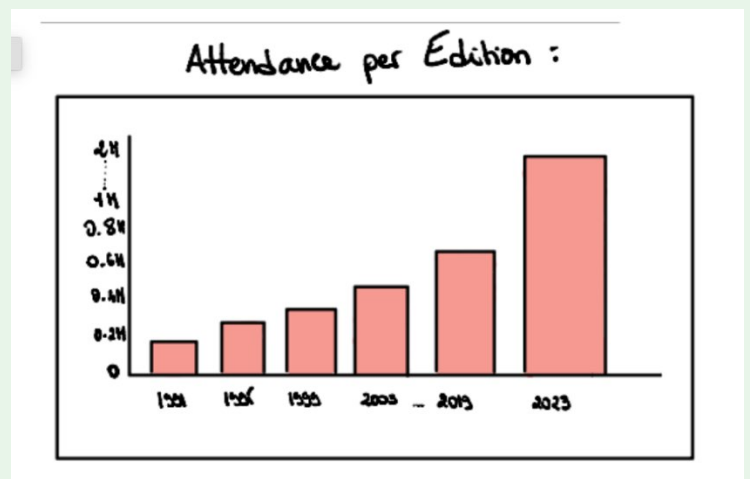
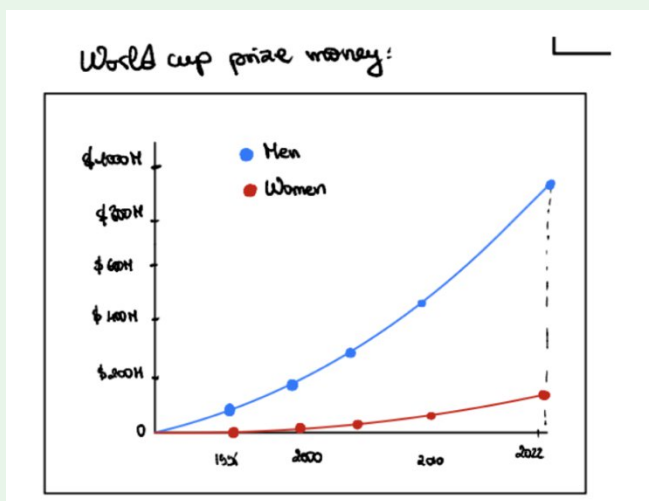
We additionally wired the map to act as a cross-filter hub: clicking any country simultaneously filters the bar chart race and the nations scatter plot, so the user can immediately pivot from geographic context to competitive history.



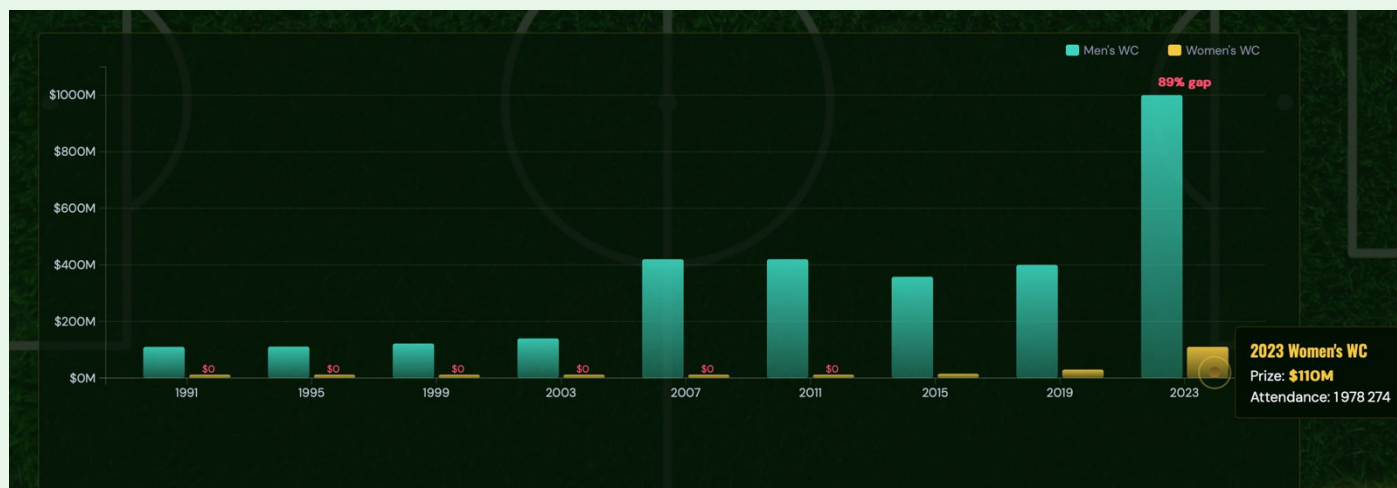
3.2 The Economics: From Two Separate Charts to One Unified Chapter

Our Milestone 2 sketches for the economics section produced two separate charts. The first was a **line chart** plotting World Cup prize money for men and women as two diverging curves from 1994 to 2022 — the men's line curving steeply upward toward \$1 billion, the women's line barely leaving zero. The second was a standalone **bar chart of attendance per World Cup edition** from 1991 to 2023, showing a consistent upward trend culminating in the 2023 record.

As we refined the economics section, two problems emerged with the original designs. The line chart obscured the most important single fact: that women received **\$0 for 24 consecutive years**. A smooth line starting at zero and barely rising reads as gradual progress — it does not convey the injustice of nothing.



We merged the two into a single **grouped bar chart** where each World Cup edition has side-by-side bars for men and women. The “\$0” annotations directly on the women’s bars from 1991 to 2011 make the absence of money impossible to ignore. The 89% gap in 2023 is annotated on the final pair of bars. The attendance story moved into four stat cards displayed above the chart — \$0 for 24 years, 89% gap today, 74% attendance growth, 1.98 million spectators in 2023 — giving the key attendance figures their own visual prominence without requiring a separate chart. This consolidation created a much more powerful, unified economic argument.



As we began working directly with the data, the dataset kept revealing angles we had not anticipated. The goalscorer records suggested a goal clock. The head-to-head match results called for a rivalry diagram. The confederation-level wins demanded a streamgraph. The 1,055 unique scorers deserved their own chapter. Each new question we asked of the data produced a chart that earned its place in the story.

By Milestone 3, we had built ten interactive visualizations, more than what we originally sketched. But the increase in volume never became an increase in complexity for the user. The scrollytelling architecture kept everything linear and guided: each chapter answers one question, then passes the reader to the next.

04 Design Decisions



Chapter 01 : Area Chart A line alone shows position; the filled area encodes volume, the reader feels the weight of 247 nations accumulating, not just a point on a scale. The animated stroke draws the history left to right on first view, turning a static fact into a felt experience.

Chapter 02 : Bar Chart Race A static ranking hides the drama of how the USA overtook Germany. Animation makes the viewer feel the debut and the rise in real time. Speed control lets users pause at key moments; the 15-team limit keeps the chart readable without sacrificing competitive depth.

Chapter 03 : Animated Map A static map shows the endpoint, not the journey. The animated year-by-year reveal lets the user watch the sport cascade outward from Europe: each nation lighting up at its exact moment of entry. The three-state color system (rose = debut, gold = active, grey = inactive) communicates events rather than values.

Chapter 04 : Chord Diagram A matrix heatmap would show the same numbers, but rivalry is fundamentally about connection, not a grid of cells. The chord makes asymmetry visible at a glance the ribbon is thicker on the dominant side. Hover isolation fades all unrelated matchups, reducing cognitive load.

Chapter 05 : Streamgraph Stacked areas with smooth curves minimize individual stream slopes, making the continental power shift readable over seven decades. The trade-off harder to read absolute values is compensated by tooltips showing exact win counts and percentages on hover.

Chapter 06 : Grouped Bar Chart A line chart was our original sketch, but a smooth curve starting at zero obscures the most important fact: women received \$0 for 24 consecutive years. Side-by-side bars with explicit "\$0" annotations make the absence of money impossible to ignore rather than a barely-visible baseline.

Chapter 07 : Force-Directed Bubble Chart Goalscorer data has no natural spatial ordering — there is no meaningful x or y axis for 1,055 players. A force simulation creates organic, non-overlapping clusters that feel alive. Bubble radius encodes goals; the physics engine handles layout at a scale no manual positioning could achieve.

Chapter 08 : Radial Goal Clock The polar form directly suggests a clock — the correct mental model for match time. The 45th and 85th minute spikes physically protrude beyond the reference circle, making the injury-time pressure pattern instantly legible without reading axis labels.

Chapter 09 : Bubble Scatter Plot Encoding goals scored as bubble radius adds a third dimension without adding another chart. Four labelled quadrants (Dominant / Rising / Experienced / Developing) give immediate context to any nation's position.

Chapter 10 : Radar Chart Five dimensions on a single polygon let two nations be compared holistically rather than through five separate charts. The dual overlay makes it immediately clear which team leads on which axis. Dropdown selects make the comparison personal — the user chooses the matchup.

05 Technical Challenges & Solutions



Building a seamless ten-chapter interactive experience using purely native technologies presented several significant hurdles.

5.1 Matching Team Names to Country Polygons

The Problem

The dataset uses plain team names “West Germany”, “Chinese Taipei”, “Ivory Coast” that do not match TopoJSON feature IDs or standard geographic lookup keys. We had 247 team names to reconcile across a 69-year span of changing borders and naming conventions.

The Solution

We built a two-layer matching strategy. The first layer is a countryAliases lookup table in JavaScript covering the most common mismatches: "usa" → "unitedstates", "chinesetaipei" → "taiwan", "ivorycoast" → "cotedivoire", "northmacedonia" → "macedonia". The second layer is a fuzzy substring matching function: after normalization (lowercase, non-letter characters stripped), if one name contains the other as a substring, it is accepted as a match. This combination handled the full range of naming inconsistencies and fails gracefully — a country that cannot be matched is left uncolored rather than crashing.

5.2 Label Collision in Dense Scatter and Bubble Charts

The Problem

Two of our visualizations — the Nations scatter plot (Chapter 09) with 30 labelled bubbles, and the Legends force bubble chart (Chapter 07) with 50+ labelled scorers — suffered from severe label overlap when using default D3 positioning. Standard approaches like fixed offsets or manual adjustments failed because the overlap pattern changed dynamically based on screen width.

The Solution

For the scatter plot, we implemented a custom **40-iteration pairwise force-repulsion algorithm**. In each iteration, every label checks its distance against every other label; if the distance falls below 28 pixels, both labels are pushed apart proportionally. After the iterations complete, connector lines are drawn from each label to its corresponding bubble if the displacement is large enough to create ambiguity. This approach is computationally heavier than a library-based solution, but gives us precise control over the visual result.

For the bubble chart, we used a simpler approach: labels are hidden unless the bubble's radius exceeds 16 pixels, and text size scales proportionally from 7pt to 13pt based on the radius. This ensures that only the most prominent scorers carry labels, keeping the chart readable.

5.3 Streamgraph Stack Order Bug

The Problem

Our initial implementation of the confederation streamgraph (Chapter 05) used `d3.stackOrderInsideOut`, which is the standard algorithm for streamgraphs. However, this order buried Europe — the historically dominant confederation — beneath several smaller groups, making the chart deeply misleading. A user reading the chart would have assumed that Asia or South America was historically dominant, which is the opposite of the truth.

The Solution

We switched to `d3.stackOrderAscending`, which places the largest band (Europe) at the visual top of the stack where it is most prominent and legible. This required two separate git commits because the first fix introduced a different rendering artifact. The experience was a clear reminder that stack order in a streamgraph is a **narrative decision**, not a technical default — it determines what story the user reads.

06 Visual Language & Architecture



6.1 Color Palette & Typography

We approached the design of “Beyond the Pitch” with the mindset of a digital sports magazine. Every visual decision was made to immerse the user in the world of women’s football, not just present data about it.

The Football Pitch Background

The most distinctive visual choice of the project is its foundation: a real football pitch photograph (bg_pitch.jpg) that covers the entire website as a fixed background. As the user scrolls through all ten chapters, the grass is always visible underneath — the data sits on the field where the game is played.

Player Photographs

The hero section opens with a rotating gallery of four real player photographs, establishing immediately that this is a story about people, not just numbers. The Legends chapter (Chapter 07) pairs the force-directed bubble chart with individual portraits of Birgit Prinz, Marta, and Abby Wambach placed directly alongside their bubbles.

- **Gold (#f5c842):**  Our primary data color, used for active states, primary data highlights, and the women’s tournament bars in the economics chart. Gold evokes achievement, and the color choice is intentional — this is the color associated with winning.
- **Teal (#3dd6c0):**  Used for contrast, the men’s tournament bars in the economics comparison, and the Asia confederation. It provides visual separation without competing with the primary gold.

For typography, we selected three complementary typefaces. **Bebas Neue** is used exclusively for large numbers and chart callouts — its tall, condensed letterforms create maximum visual impact for statistics like “247 nations” or “\$110M.” **Oswald** handles navigation, section titles, and axis labels — condensed enough to fit in tight spaces while remaining authoritative. **DM Sans** handles all body text, tooltips, and descriptions — clean and highly readable at small sizes.

6.2 Frontend Architecture

By building exclusively with native HTML, CSS, and JavaScript, without React, Vue, or any component framework, we ensured our site remained fast, lightweight, and fully deployable as a static file on GitHub Pages.

All data is preprocessed in Python and exported as JSON, then bundled into a single bundle.js file that is loaded once at page start. This means the visualization works fully offline, requires no backend infrastructure, and loads in under one second on a standard connection.

The D3.js visualizations are all written as self-contained functions in charts.js, each accepting a container element and a data object. A separate crossfilter.js module manages the shared state that links the world map to the bar chart race and the nations scatter plot. The scroll system in interactions.js uses the native IntersectionObserver API to trigger chart initialization and entrance animations as each chapter enters the viewport, avoiding the performance cost of initializing all ten charts at page load.

07 Peer assessment & Conclusion



7.1 Team Contributions

This project was a collaborative effort, with each team member bringing distinct expertise and responsibility.

Imane Oujja (344332): Imane was responsible for the visual design system and four core visualizations: the History Area Chart (Chapter 01), the Rivalries Chord Diagram (Chapter 04), the Economics Grouped Bar Chart (Chapter 06), and the Radar Compare Chart (Chapter 10). She defined the color palette, selected the typography stack, and designed the overall page layout. She also led the Python EDA phase, compiled the supplementary prize money and attendance data from FIFA and Wikipedia, and authored the Milestone 1 written report.

Shyamala Vasireddy (423053): Shyamala handled the data story and two technically complex visualizations: the Legends Force-Directed Bubble Chart (Chapter 07) and the Goal Patterns Radial Clock and Activity Chart (Chapter 08). She built the tooltip positioning system used across all ten charts. She also built the country alias lookup table for the 49 edge cases, conducted the goalscorer data completeness audit, implemented the `has_data` coverage flag, and produced the two-minute screencast.

Rim Abkari (344316): Rim led the full technical architecture of the project. She built the Python data preprocessing pipeline, designed the data bundle structure, and implemented four of the ten D3 visualizations: the Bar Chart Race (Chapter 02), the World Map (Chapter 03), the Streamgraph (Chapter 05), and the Nations Scatter Plot (Chapter 09). She also built the cross-filtering system linking the map to downstream charts, the complete scroll interaction system, the custom cursor, the loading screen, and the GitHub Pages deployment. Rim was responsible for final performance optimization and for fixing the streamgraph stack order bug.

(While tasks were divided for efficiency, all major design decisions, architectural pivots, and narrative directions were discussed and agreed upon collectively by the team.)

7.2 Conclusion & Future Work

“*Beyond the Pitch*” successfully transforms 69 years of sports data into an accessible, human-centered story. We proved that data visualization is not just about drawing charts — it is about guiding a user toward a revelation. The revelation here is quiet but powerful: a sport that could not legally be played in England in 1970 now has 238 participating nations, 2 billion television viewers, and a prize fund that has grown from zero to \$110 million in three decades.

If we had additional time to expand this project, we would explore two directions. First, sub-national data: state-by-state or province-level breakdowns to reveal inequalities within countries: for example, how women’s football participation varies by region within the United States or Brazil. Second, a live data connection: an API pipeline that automatically updates the visualizations with new match results, allowing the story to continue growing in real time.

We are proud of what “*Beyond the Pitch*” became, not just a course deliverable, but a piece of interactive journalism that tells a story worth telling.